Mohammad Tanzil Idrisi

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Summary

AI systems engineer and founder building production ML infrastructure, distributed systems, and developer platforms. Shipped large-scale ML systems at Meta, built low-latency infrastructure serving 10M+ daily queries, and published research across efficient inference, multimodal AI, and medical imaging.

Education

Beloit College — **December 2025**

Bachelor of Science in Computer Science and Mathematics — *Beloit, Wisconsin*

- **GPA: 3.9/4.0**, **Scholarship & Awards:** Presidential Scholarship, Dean's List, Google Software Engineer Fellow

Experience

Convertive (Previously Outhad) (Website) — **Sept 2024 – Present**

Co-founder & Chief Technical Officer — *Madison, Wisconsin*

- Raised $250K pre-seed and built an AI marketing platform with distributed backend infrastructure
- Scaled Convertive to $1M+ revenue while owning growth, architecture, infrastructure, and reliability
- Processed millions of web, session, cart, and conversion events with Go/C++/Rust pipelines sustaining 99.9% uptime
- Built low-latency C++ decisioning engine for millisecond-scale real-time personalization and marketing workflows
- Served 10M+ daily analytics queries using PostgreSQL/Redis replication, caching, and query optimization
- Scaled 50K+ marketing workflows/hour with TypeScript services, PostgreSQL, Redis, and Kubernetes HPA
- Designed fault-tolerant ETL pipelines with retries, audit logs, rate limits, validation, and recovery paths
- Improved data observability with tracing, metrics, logs, anomaly checks, and automated failure detection

Shopify — **May 2025 – Aug 2025**

Software Engineer Intern — *San Francisco, California*

- Built scalable ads auction system in Go for handling 10M+ daily requests at p95 under 120ms
- Maintained 99.9% auction availability across multi-AZ Kubernetes deployment and failover
- Shipped feature-flag service powering checkout experiments with automated rollout controls
- Reduced PostgreSQL load 40% with Redis caching for sub-millisecond metadata lookups
- Processed billions of events with Airflow pipelines, distributed tracing, and automated recovery

DAAD German Academic Exchange Service — **May 2024 – Aug 2024**

DAAD Rise Scholar - Undergraduate Researcher — *Hamburg, Germany*

- Developed and evaluated noise-aware 3D U-Net models in PyTorch, improving PET image quality 15%
- Presented model architecture, evaluation methodology, and results at IEEE NSS Medical Imaging Conference 2024

Carnegie Mellon University, School of Computer Science — **Aug 2023 – Jan 2024**

Machine Learning Research Intern — *Remote*

- Built a PyTorch GAN pipeline reconstructing 3D faces from 2D images for surgical planning
- Fine-tuned on CMU Multi-PIE, achieved 90.2% LFW accuracy, and mapped 68 landmarks to FLAME meshes

Meta — **May 2023 – Aug 2023**

Software Engineer Intern — *Menlo Park, CA*

- Built Vision Transformer optimizations for large-scale video understanding workloads across Meta production systems
- Designed adaptive token sampling in PyTorch and integrated it into Reels video understanding pipelines
- Reduced transformer inference latency and FLOPs by 25%, cutting production compute costs by 10%.

FractalXR — **Feb 2020 – April 2021**

Founder — *Mumbai, India*

- Turned down Caltech at 19 to build FractalXR during COVID's shift to remote collaboration
- Built FractalXR from scratch as an AR/VR collaboration platform for spatial virtual meetings
- Led product and engineering through acquisition by BoxAlly.

Publications

Paired Replacement: Differential Cache Maintenance for Dynamic Sparse and Mixture-of-Experts Inference
AIML Systems Conference 2026 (Accepted) *Oct, 2026*

Timeline-Weighted Contextual Bandits for Real-Time E-Commerce Intervention: Representation, Benchmarks, and an Honest Off-Policy Study
Zenodo record *June 12, 2026*

C-PETAL-3D: Classification Driven Progressive Elimination of Noise Towards Accurate Ultra Low Dose PET Images Using 3D U-Nets
IEEE NSS MIC RTSD 2024 *Oct 28, 2024*

Talks

ContextKit: Multimodal Graph Memory Architecture for AI
National Conference of Undergraduate Research 2024, Pittsburg, PA *April 7, 2025*

Contextkit: Memory layer for long running agent
Mid-States Consortium Undergraduate Research Symposium 2023, U. Chicago *Oct 23, 2024*

Efficient Hyperparameter Optimization through Sparse Sampling and Robust Tensor PCA
National Conference of Undergraduate Research 2024, Miami, Florida *Apr 10, 2024*

Quantization Optimization: Autoencoders and Lloyd-Max for Data Compression
Mid-States Consortium Undergraduate Research Symposium 2023, U. Chicago *Oct 23, 2023*

Pooling Convolutional Capsule Network
National Conference of Undergraduate Research 2023, University of Wisconsin-Eau Claire *April 12 2023*

Pooling Convolutional Capsule Network
Mid-States Consortium Undergraduate Research Symposium 2022, U. Chicago *Oct 23, 2022*

Projects

ControlDB: Transactional State Engine for AI Agents | C++, Python, Raft, RocksDB, SQL **(Github)** **Dec 2025**
- Built a C++ SQL state engine for durable agent memory, tool logs, replay, and audit trails
- Implemented Raft replication, RocksDB storage, transactions, and failover tests for agent workflows

ContextKit: Intelligent Memory Layer for LLM Apps | Python, Neo4j, Qdrant **(PyPI)** **Nov 2025**
- Cut context cost 99.6% and improved long-context accuracy 36% with Python memory engine
- Built Neo4j graph memory with PII/PHI detection, encryption, and multi-LLM timeline reasoning

PromptKernel | Cloudflare Workers, KV, Vectorize, Durable Objects, TypeScript | **Article / Live Demo** **Mar 2026**
- Built a Cloudflare-native semantic LLM cache with approximately 8 ms exact hits and 25 ms semantic hits
- Reached a 97.5% cache hit rate and approximately 97% provider-cost reduction across 8,000 messages
- Published a technical article and live demo explaining architecture, benchmarks, and developer integration

Franz: Functional Programming Language | C, LLVM **2020**
- Developed a new functional, prototype-oriented programming language
- Built a 96K-line C compiler targeting LLVM with C/Rust-like performance via aggressive optimization passes

SynergyML | Python, PyTorch, Hugging Face, OpenCV, TimeSformer, Wav2Vec2 | **GitHub** **Nov 2024**
- Built a multimodal framework combining language, image, video, and audio models through a unified Python API
- Integrated Hugging Face Transformers with classical ML workflows and reusable developer-facing examples

IonicTrace | Rust, C/C++, Python | **(Github)** **Dec 2025**
- Built Rust tracing system with eBPF/XDP and io_uring, achieving 49.8 ns checks and 100M+ packets/sec
- Achieved real-time event correlation under 1 ms with <100 ns network latency overhead in production

Technical Skills

Languages: Python, TypeScript/JavaScript, Go, C++, Rust, SQL

AI/ML: PyTorch, Hugging Face Transformers, Sentence Transformers, OpenCV, scikit-learn, model fine-tuning, inference, vector search, multimodal ML, Workers, KV, Vectorize, Durable Objects

Infrastructure: Kubernetes, Docker, Terraform, AWS, GCP, PostgreSQL, Redis, Kafka, Airflow, distributed tracing, REST APIs, gRPC, developer tooling